**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 074 · Random Number Generation

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What does WHERE do in an SQL query?

Q2. What does a MAC address identify?
A. A website's domain name
B. A network interface
C. A user's password
D. A file's folder

Q3. “A firewall confirms that every sentence on every website is true.” Correct this. Explain why.

Q4. Data must reach another network in the right order. “TCP chooses fonts; IP draws images.” Correct this. Explain each protocol's job.

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# I Do · Read new knowledge

## Key vocabulary

Random numbers — random(low,high) produces a value inside the inclusive range.

Integer range — random(1,6) can return any whole number from 1 to 6.

Unpredictability — Successive random calls can produce different values; outputs are not fixed.

Simulation — Random values can model dice, cards or game events without physical devices.

Algorithm — A defined set of processing instructions.

Testing — Checking expected outcomes against actual program behaviour.

## New learning

Random numbers: random(low,high) produces a value inside the inclusive range. In the dice game, identify where this technique applies.

Integer range: random(1,6) can return any whole number from 1 to 6. In the quiz bonus round, identify where this technique applies.

Unpredictability: Successive random calls can produce different values; outputs are not fixed. In the board-game move, identify where this technique applies.

Simulation: Random values can model dice, cards or game events without physical devices. In the random test selector, identify where this technique applies.

## Worked example

A dice game uses random numbers and integer range. Explain what each contributes. [4 marks]

1. Identify the program's purpose and the pieces of data needed.

2. Random numbers [1]; random(low,high) produces a value inside the inclusive range. [1].

3. Integer range [1]; random(1,6) can return any whole number from 1 to 6. [1].

4. Each feature needs a correct technical fact plus an explanation.

# We Do · Practise with guidance

### Guided 1

A quiz bonus round needs random numbers. Explain the purpose. [2 marks]

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Worked method: Random numbers [1]; random(low,high) produces a value inside the inclusive range. [1].

### Guided 2

Explain integer range for a board-game move. [2 marks]

Worked method: Integer range [1]; random(1,6) can return any whole number from 1 to 6. [1].

### Guided 3

When programming a random test selector, explain why unpredictability and simulation matter. [4 marks]

Worked method: Unpredictability [1]; Successive random calls can produce different values; outputs are not fixed. [1]. Simulation [1]; Random values can model dice, cards or game events without physical devices. [1].

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# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Write one OCR ERL instruction to generate an integer from 1 to 6 inclusive.

```
roll = ...
print(roll)
```

write

### Question 1

State TWO relevant ideas or operations from this topic. [2 marks] Source: Original OCR J277-style teaching question; not an official past paper.

### Question 2

Explain integer range and unpredictability in a virtual spinner. [4 marks] Source: Original OCR J277-style teaching question; not an official past paper.

### Question 3

A programmer makes a sports draw. Explain the roles of random numbers, unpredictability and simulation. [6 marks] Source: Original OCR J277-style teaching question; not an official past paper.

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## Exit Ticket · Answer before checking

random(low,high) produces a value inside the inclusive range.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

## Page 1 onwards · Do Now

Q1. It selects rows that meet a condition.

Q2. B — A network interface

Q3. It uses rules to allow or block network traffic.

Q4. TCP supports reliable delivery in order. IP handles addressing and routing between networks. Each protocol has its own job; encryption does not prove a site is honest.

## I Do / We Do pages · Guided practice

Q1. Random numbers [1]; random(low,high) produces a value inside the inclusive range. [1].

Q2. Integer range [1]; random(1,6) can return any whole number from 1 to 6. [1].

Q3. Unpredictability [1]; Successive random calls can produce different values; outputs are not fixed. [1]. Simulation [1]; Random values can model dice, cards or game events without physical devices. [1].

## You Do pages · Code First

```
roll = random(1,6)
print(roll)
```

Expected result: Any whole number from 1 to 6 inclusive (not a fixed output)

## You Do pages · Independent questions

Q1. Random numbers [1]; Integer range [1].

Q2. Integer range [1]; random(1,6) can return any whole number from 1 to 6. [1]. Unpredictability [1]; Successive random calls can produce different values; outputs are not fixed. [1].

Q3. Random numbers [1]; random(low,high) produces a value inside the inclusive range. [1]. Unpredictability [1]; Successive random calls can produce different values; outputs are not fixed. [1]. Simulation [1]; Random values can model dice, cards or game events without physical devices. [1].

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### Exit Ticket · Model answer

True

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**